Homework 4
Statistics 2103

1. In this question you will explore the relationship between the barometric pressure at the center of a hurricane and the hurricane's maximum wind speed. The file `hurricanes.csv` contains the following variables for 62 hurricanes that made landfall in the US in the last 25 years:

   pressure: central pressure in millibars
   windspeed: maximum wind speed in knots

   (a) Find the least squares estimator of the slope and intercept of a regression model for this data, using `pressure` as the predictor (X) and `windspeed` as the response (Y).

   (b) Make a scatter plot of the data. Display the estimated regression line on the scatter plot.

   (c) Describe the interpretation of the slope and intercept estimated in part (a).

   (d) What are the units of the slope and the intercept?

   (e) According to the estimated model, on average what maximum wind speed do we expect for hurricanes with a central pressure of 945 millibars?

   (f) According to the estimated model, on average what maximum wind speed do we expect for hurricanes with a central pressure of 750 millibars? Do you think this is a reasonable prediction to make?

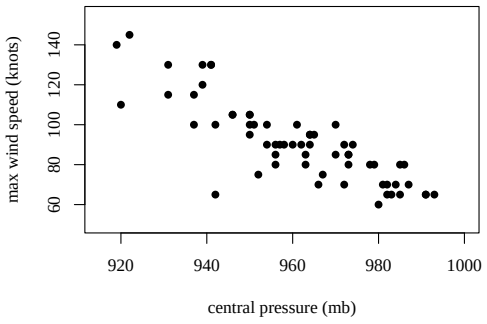
   (g) Compute the average of the observed central pressure values, $\bar{x}$. What max wind speed does the model predict for a storm with central pressure equal to $\bar{x}$? Compare this value to the average of the observed max wind speed values, $\bar{y}$.

   (h) Compute the standard deviation of the observed central pressure values (s_x). For hurricanes whose central pressures differ by that amount, what is the difference in predicted max wind speeds?

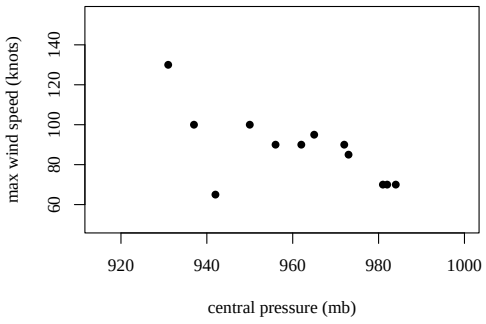
   (i) How does the difference in predicted wind speeds in part (h) compare to the standard deviation of the observed max wind speed values (s_y)?

   (j) Hurricane Harvey had a central pressure of 937 millibars and a max wind speed of 115 knots. What is the residual for that data point?

2. The scatter plots below show two sample data sets. Which sample would you expect to provide a more precise estimate of the true underlying linear relationship between X and Y? Give a brief justification.

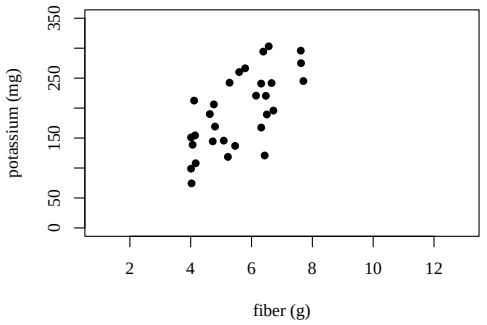


(a)

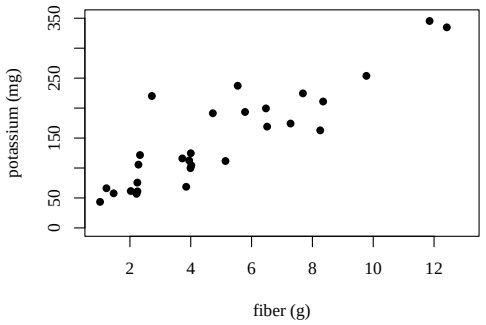


(b)

3. The scatter plots below show two sample data sets. Which sample would you expect to provide a more precise estimate of the true underlying linear relationship between X and Y? Give a brief justification.

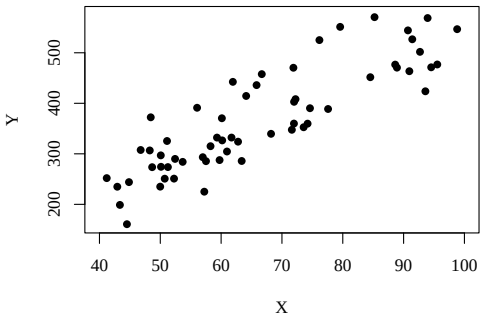


(a)

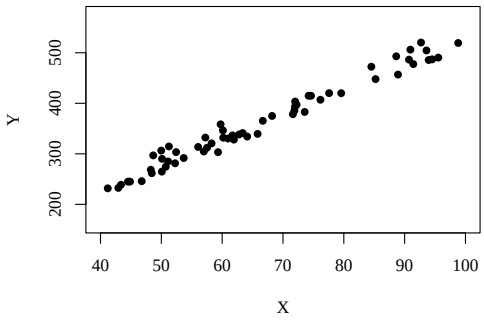


(b)

4. The scatter plots below show two sample data sets. Which sample would you expect to provide a more precise estimate of the true underlying linear relationship between X and Y? Give a brief justification.



(a)



(b)